AI-Driven Analytics for Energy-Efficient and Cost-Aware Data Center Planning

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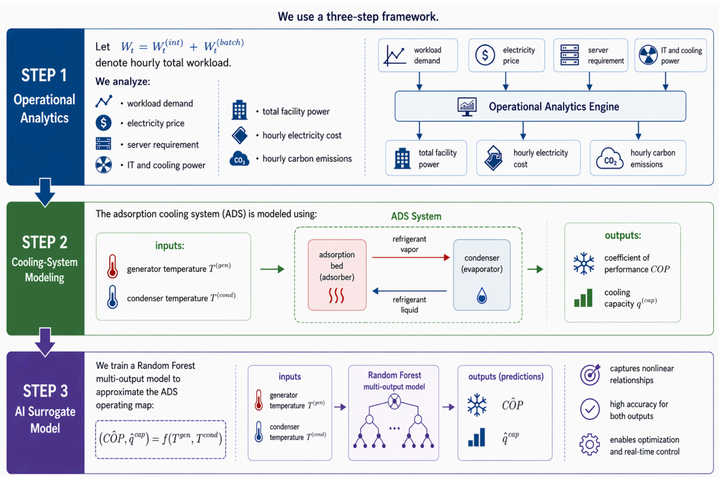
I. Background

- Data centers consume large amounts of energy for both computing and cooling, making energy efficiency a major challenge.
- Operators must balance workload demand, electricity cost, cooling needs, and carbon emissions while maintaining reliable performance.
- This study combines descriptive analytics and AI modeling to support efficient, cost-aware, and sustainable data center planning.

II. Dataset Overview

- The dataset includes hourly workload demand, electricity prices, server parameters, and ADS cooling-system states.
- These variables support integrated analysis of both computing demand and cooling-system performance under changing operating conditions.
- The dataset supports analysis of:
  - power demand
  - operating cost
  - carbon emissions
  - cooling performance
- It also provides the foundation for AI-based modeling of ADS efficiency and cooling capacity.

III. Methodology



IV. Key Equations

$$\text{Electricity Cost} = \text{Energy Consumption} \times \text{Electricity Price} \quad (1)$$

$$\text{Carbon Emissions} = \text{Energy Consumption} \times \text{Carbon Intensity} \quad (2)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (3)$$

V. Workload and Electricity Price

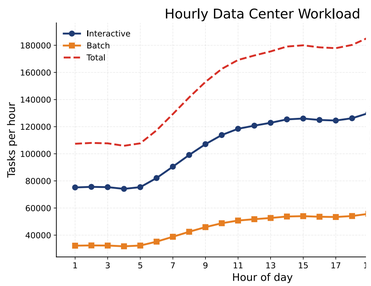


Figure 1. Daily workload pattern.

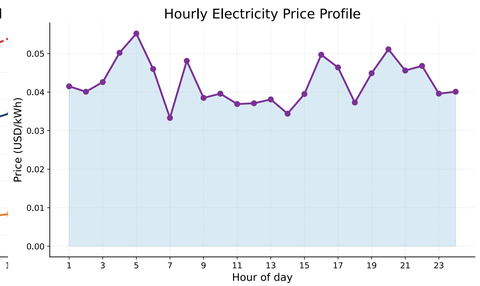


Figure 2. Hourly electricity price.

Key Insight: Workload and price peaks do not fully align, creating opportunities for cost-aware scheduling of flexible demand.

VI. Power, Cost, and Carbon Results

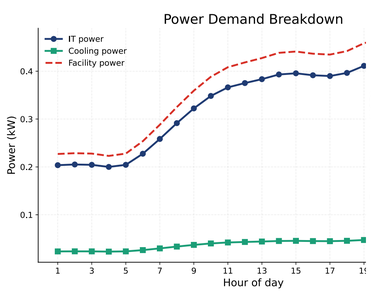


Figure 3. Facility power breakdown.

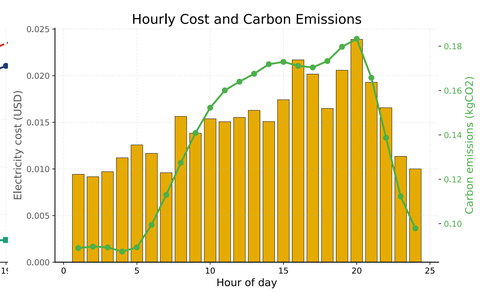


Figure 4. Cost and carbon emissions.

Observation: Higher workload increases both IT and cooling demand, which raises cost and carbon emissions.

VII. Cooling-System Performance Maps

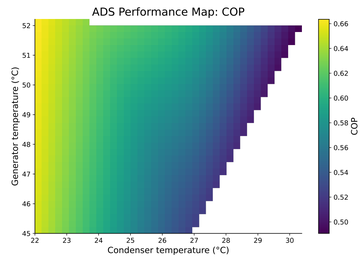


Figure 5. ADS COP map.

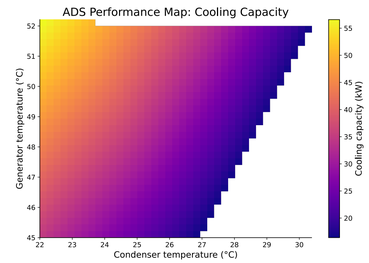


Figure 6. ADS capacity map.

Key Finding: Maximum efficiency and capacity occur in different operating regions.

VIII. AI Model Validation

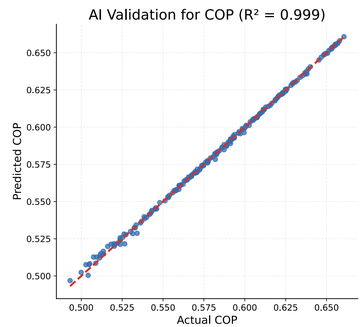


Figure 7. AI validation for COP.

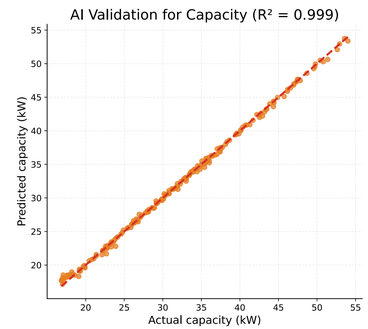


Figure 8. AI validation for capacity.

Model Result: The Random Forest surrogate predicts both COP and cooling capacity accurately and can support optimization and control.

IX. Conclusion

- Workload and price profiles reveal scheduling opportunities
- Facility power, cost, and emissions vary substantially over time
- ADS performance is strongly nonlinear
- A Random Forest surrogate accurately predicts COP and cooling capacity